

E-Commerce US Dataset Analysis

Report 3: Data Quality Assessment

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1. Scope

- Assessment covers both integrated tables: master_orders (10,000 rows × 36 cols) and items_enriched (21,800 rows × 24 cols).
- Five quality dimensions checked: completeness, uniqueness, data types, business rules, and consistency.

2. Completeness — Missing Values

2.1 master_orders

Column	Null Count	Null %	Root Cause
order_delivered_carrier_date	668	6.68%	Canceled orders — no carrier date assigned
order_delivered_customer_date	668	6.68%	Canceled orders — never reached customer
avg_review_score	668	6.68%	Canceled orders — no review submitted
n_reviews	668	6.68%	Canceled orders — same 668 records
All other columns	0	0.00%	—

- All 668 null records are canceled orders — the nulls are structurally expected, not data entry errors.
- No treatment needed; these rows are flagged with is_delivered = 0 and has_review = 0 during cleaning.

2.2 items_enriched

- Zero missing values across all 21,800 rows and 24 columns.
- Product and seller dimensions fully populated — no orphaned item rows.

3. Uniqueness — Duplicate Check

Table	Check	Result	Status
master_orders	Full-row duplicates	0	PASS
master_orders	Duplicate order_id values	0	PASS
items_enriched	Full-row duplicates	0	PASS
items_enriched	Duplicate (order_id, item_id)	0	PASS

4. Data Type Integrity

Column	Source Type	Required Type	Issue
order_purchase_timestamp	str (object)	datetime64	String storage in source system
order_approved_at	str (object)	datetime64	String storage in source system
order_estimated_delivery_date	str (object)	datetime64	String storage in source system
order_delivered_carrier_date	str (object)	datetime64	String storage in source system
order_delivered_customer_date	str (object)	datetime64	String storage in source system
shipping_limit_date (items)	str (object)	datetime64	String storage in source system

- All 6 date columns converted with pd.to_datetime() in the cleaning notebook.
- All numeric columns (price, freight, coordinates, review_score) already in correct types pre-integration.

5. Business Rule Violations

Rule	Violations	% of Rows	Action
total_payment > 0	0	0.00%	Pass — no action
total_price > 0	0	0.00%	Pass — no action
n_items >= 1	0	0.00%	Pass — no action
review_score in {1,2,3,4,5}	0	0.00%	Pass — no action

price >= cost (products)	0	0.00%	Pass — no action
customer_age in [18, 100]	804	8.04%	Replaced with median (48)

6. Outlier Detection (IQR Method)

Column	Lower Fence	Upper Fence	Outliers	%	Action
total_payment	-\$1,675.90	\$3,857.70	225	2.25%	Retained; flagged is_high_value
total_price	-\$1,761.10	\$3,481.00	221	2.21%	Retained
n_items	-2.0	6.0	0	0.00%	Pass
customer_age	3.0	67.0	35	0.35%	Covered by age treatment
price (items)	-\$947.10	\$1,746.40	863	3.96%	Retained (valid premium tier)
product_weight_g	-3,945 g	8,135 g	2,297	10.54%	Retained (catalog diversity)

7. Consistency Checks

Check	Result	Status
Each customer_city maps to exactly one customer_state	Perfect 1-to-1 mapping	PASS
Each seller_city maps to exactly one seller_state	Perfect 1-to-1 mapping	PASS
customer_segment values within {Consumer, Corporate, VIP}	All values valid	PASS
order_status values within {delivered, canceled}	Only 2 values present	PASS
payment_type values within defined set (7 types)	All within expected set	PASS
product_category values within defined set (7 categories)	All within expected set	PASS

8. Referential Integrity

- All order_id values in items_enriched exist in master_orders — zero orphaned items.
- All customer_id values in orders exist in the customers table — zero orphaned orders.
- All product_id and seller_id values in order_items resolve to the products and sellers tables.
- Geolocation joins: 849 distinct zip codes from customers all map to geolocation entries.

9. Summary Scorecard

Dimension	Finding	Status
Completeness	6.68% structural nulls (canceled orders); 0% unexpected nulls	Good
Uniqueness	Zero duplicate rows or key collisions in both tables	Excellent
Data Types	6 date columns stored as strings — fixed in cleaning	Fixed
Business Rules	804 age violations treated by median imputation	Treated
Consistency	All categorical values within expected domain sets	Excellent
Referential Integrity	Zero orphaned rows across all foreign key relationships	Excellent
Outliers	2.2% high-value orders retained and flagged	Noted